Experiment 3: Online Bus Ticket Booking System

Aim

To design and develop an online bus ticket booking website using HTML, CSS and JavaScript in which passengers can register, log in, search buses by route, date and time, view available seats, select seats, enter passenger details, pay through a payment gateway, view booking history, cancel tickets and post comments on a bus; and the administrator can manage bus categories, bus details, users, news and view reports.

Algorithm

1. Start and display the login page (Login, Forgot Password and Registration forms).
2. Registration: validate name, email, 10-digit phone, password (minimum 6 characters) and unique username; store the passenger.
3. Login: verify username and password for the selected role. Forgot password: if username and email match, set a temporary password and notify by email. Change password: verify the old password, then check that the new and confirm passwords match.
4. Passenger - Search: read from, to, date and time of day (morning / afternoon / evening / night) and list matching buses with fare and seats left = total seats - seats already booked for that bus and date.
5. Passenger - Select seats: show the seat layout; booked seats are disabled; allow up to 6 seats; total = seats x fare.
6. Passenger - Enter name and age for every selected seat; validate all fields.
7. Payment: validate the 16-digit card number, name on card, expiry (MM/YY) and 3-digit CVV; re-check that the seats are still free; save the ticket and send a confirmation email.
8. Passenger - View booking history; cancel a ticket (status Cancelled, seats released, 80% refund); post and read comments on a bus; read news.
9. Admin - Add or delete bus categories (Volvo, Sleeper Coach, Coach).
10. Admin - Add, update or delete buses (name, category, route, time, fare, seats, photo, description); a bus with upcoming bookings cannot be deleted.
11. Admin - Manage users, publish or delete news, and delete comments.
12. Admin - Generate the report: total buses, users, confirmed and cancelled bookings, revenue, and bookings, seats sold and revenue per bus.
13. Logout returns to the login page. Stop.

Program (HTML, CSS and JavaScript) - bus.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1, viewport-fit=cover">
<title>Online Bus Ticket Booking</title>
<style>
:root{box-sizing:border-box;padding-top:env(safe-area-inset-top,0px);padding-bottom:env(safe-area-inset-bottom,0px);
--bg:#eef3f6;--card:#fff;--tx:#1b2a33;--mut:#64747e;--ac:#0b7285;--acx:#fff;--bd:#d3dde3;--bad:#c0392b;--ok:#1e8449;--
    sel:#f59f00}
@media (prefers-color-scheme:dark){:root:not([data-theme="light"]){--bg:#10181c;--card:#1a252b;--tx:#e6eef2;--mut:#93a4ae;--
    ac:#3bb5cc;--acx:#06181d;--bd:#31434c;--bad:#ef7b6e;--ok:#58c183;--sel:#fcc419}}
html{scroll-padding-top:env(safe-area-inset-top,0px)}
*{box-sizing:border-box}
body{margin:0;background:var(--bg);color:var(--tx);font-family:system-ui,Segoe UI,Arial,sans-serif;line-height:1.45}
header{background:var(--ac);color:var(--acx);padding:12px 16px}
header h1{margin:0 0 8px;font-size:1.15rem}
header nav{display:flex;flex-wrap:wrap;gap:4px}
header button{background:transparent;color:inherit;border:1px solid transparent;padding:5px 10px;border-
    radius:5px;cursor:pointer;font-size:.88rem}
header button.on,header button:hover{border-color:currentColor}
main{max-width:980px;margin:16px auto;padding:0 12px}
.card{background:var(--card);border:1px solid var(--bd);border-radius:10px;padding:16px;margin-bottom:14px}
button.b{background:var(--ac);color:var(--acx);border:0;border-radius:6px;padding:8px 14px;cursor:pointer;font-size:.9rem}
button.a{background:transparent;color:var(--tx);border:1px solid var(--bd)}
button.s{padding:3px 9px;font-size:.8rem}
input,select,textarea{padding:8px;border:1px solid var(--bd);border-radius:6px;background:var(--bg);color:var(--tx);font-
    size:.9rem;max-width:100%}
label{font-size:.85rem;color:var(--mut);display:flex;flex-direction:column;gap:3px}
.row{display:flex;gap:10px;flex-wrap:wrap;align-items:flex-end;margin-bottom:10px}
.tw{overflow-x:auto}
table{border-collapse:collapse;width:100%;font-size:.88rem}
th,td{text-align:left;padding:7px 9px;border-bottom:1px solid var(--bd);white-space:nowrap}
th{color:var(--mut)}
h2{margin:0 0 10px;font-size:1.05rem}
.bad{color:var(--bad);font-weight:700}.ok{color:var(--ok);font-weight:700}.mut{color:var(--mut);font-size:.85rem}
.login{max-width:360px;margin:0 auto}.login input,.login select{width:100%;margin-bottom:8px}
.seats{display:grid;grid-template-columns:repeat(5,52px);gap:8px;margin:10px 0}
.seat{height:44px;border-radius:6px;border:1px solid var(--bd);background:var(--bg);color:var(--tx);cursor:pointer}
.seat.x{background:var(--bd);color:var(--mut);cursor:not-allowed;text-decoration:line-through}
.seat.on{background:var(--sel);color:#222;border-color:var(--sel);font-weight:700}
.gap{visibility:hidden}
.ticket{border:2px dashed var(--ac);padding:12px;border-radius:8px}
</style>
</head>
<body>
<header><h1>■ Online Bus Ticket Booking</h1><nav id="nav"></nav></header>
<main id="app"></main>
<script>
const $=id=>document.getElementById(id);
const esc=s=>String(s).replace(/[&<>"]/g,c=>({'&':'&amp;','<':'&lt;','>':'&gt;','"':'&quot;'}[c]));
const today=()=>new Date().toISOString().slice(0,10);
const add=(d,n)=>{const x=new Date(d);x.setDate(x.getDate()+n);return x.toISOString().slice(0,10)};
const msg=t=>alert(t);
let nid=100;
const D={
 admin:{u:'admin',p:'admin123'},
 users:[{u:'guest',p:'guest123',name:'Anita Rao',email:'anita@mail.com',phone:'9876543210'}],
 cats:['Volvo','Sleeper Coach','Coach'],
 buses:[
  {id:1,name:'KPN Express',cat:'Volvo',from:'Chennai',to:'Bangalore',time:'22:00',fare:900,seats:20,desc:'AC multi-axle Volvo,
    charging point, reclining seats.',photo:''},
  {id:2,name:'SRS Night Rider',cat:'Sleeper Coach',from:'Chennai',to:'Bangalore',time:'21:30',fare:750,seats:15,desc:'Non-AC
    sleeper with blankets and water bottle.',photo:''},
  {id:3,name:'TNSTC Deluxe',cat:'Coach',from:'Chennai',to:'Madurai',time:'07:00',fare:500,seats:20,desc:'Semi-sleeper day
    coach.',photo:''},
  {id:4,name:'Orange Volvo',cat:'Volvo',from:'Bangalore',to:'Chennai',time:'06:30',fare:850,seats:20,desc:'AC Volvo with
    snacks.',photo:''}],
 bk:[{id:1,u:'guest',bus:1,date:add(today(),3),seats:[3,4],pax:[{n:'Anita Rao',a:28},{n:'Ravi
    Rao',a:31}],amt:1800,st:'Booked'}],
 cm:[{id:1,bus:1,u:'guest',text:'Very comfortable journey, on time.',date:today()}],
 news:[{id:1,text:'Festival special: 10 extra buses to Madurai from 1 November.',date:today()}]
};
let role=null,me=null,view='login',q={},flow=null,cmBus=null,editBus=null;

const go=v=>{view=v;flow=null;cmBus=null;render()};
const bus=id=>D.buses.find(b=>b.id==id)||{name:'(deleted)',from:'',to:'',fare:0};
const taken=(id,date)=>D.bk.filter(b=>b.bus==id&&b.date==date&&b.st=='Booked').flatMap(b=>b.seats);
const period=t=>{const h=+t.slice(0,2);return h>=5&&h<12?'Morning':h>=12&&h<17?'Afternoon':h>=17&&h<21?'Evening':'Night'};

function render(){
 let n=[];
 if(role=='user')n=[['search','Search Bus'],['my','My Bookings'],['news','News'],['pw','Change Password']];
 if(role=='admin')n=[['cats','Categories'],['buses','Manage
    Buses'],['users','Users'],['anews','News'],['acm','Comments'],['rep','Report'],['pw','Change Password']];
 n.push(me?['out','Logout ('+esc(me.u)+')']:['login','Login / Register']);
 $('nav').innerHTML=n.map(x=>`<button class="${x[0]==view?'on':''}"
    onclick="${x[0]=='out'?'logout()':`go('${x[0]}')`}">${x[1]}</button>`).join('');
 const
    V={login:loginV,search:searchV,my:myV,news:newsV,pw:pwV,cats:catsV,buses:busesV,users:usersV,anews:anewsV,acm:acmV,rep:repV}
```

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    ;
 $('app').innerHTML=V[view]();
}
function logout(){role=me=null;go('login')}

/* ---------- login, register, forgot / change password ---------- */
function loginV(){
 return `<div class="login"><div class="card"><h2>Login</h2><select id="lr"><option value="user">Passenger</option><option
    value="admin">Administrator</option></select>
 <input id="lu" placeholder="Username"><input id="lp" type="password" placeholder="Password"><button class="b"
    onclick="login()">Login</button>
 <p class="mut">Demo: guest / guest123 | admin / admin123</p></div>
 <div class="card"><h2>Forgot Password</h2><input id="fu" placeholder="Username"><input id="fe" placeholder="Registered
    email"><button class="b a" onclick="forgot()">Reset Password</button></div>
 <div class="card"><h2>Register</h2><input id="rn" placeholder="Full name"><input id="re" placeholder="Email"><input id="rh"
    placeholder="Phone (10 digits)">
 <input id="ru" placeholder="Username"><input id="rp" type="password" placeholder="Password (min 6)"><button class="b"
    onclick="reg()">Register</button></div></div>`;
}
function login(){
 const r=$('lr').value,u=$('lu').value.trim(),p=$('lp').value;
 const f=r=='admin'?(D.admin.u==u&&D.admin.p==p?D.admin:null):D.users.find(x=>x.u==u&&x.p==p);
 if(!f)return msg('Invalid username or password');
 role=r;me=f;go(r=='admin'?'buses':'search');
}
function reg(){
 const
    o={name:$('rn').value.trim(),email:$('re').value.trim(),phone:$('rh').value.trim(),u:$('ru').value.trim(),p:$('rp').value};
 if(!o.name||!o.u)return msg('Please fill all fields');
 if(!/^\S+@\S+\.\S+$/.test(o.email))return msg('Enter a valid email');
 if(!/^\d{10}$/.test(o.phone))return msg('Phone must be 10 digits');
 if(o.p.length<6)return msg('Password must have at least 6 characters');
 if(D.users.some(x=>x.u==o.u))return msg('Username already taken');
 D.users.push(o);msg('Registered successfully. Please login.');
}
function forgot(){
 const u=D.users.find(x=>x.u==$('fu').value.trim()&&x.email==$('fe').value.trim());
 if(!u)return msg('No account matches that username and email');
 u.p='temp1234';msg('A temporary password (temp1234) has been sent to '+u.email+'. Please change it after login.');
}
function pwV(){
 return `<div class="card login"><h2>Change Password</h2><input id="op" type="password" placeholder="Old password"><input
    id="np" type="password" placeholder="New password (min 6)">
 <input id="cp" type="password" placeholder="Confirm new password"><button class="b" onclick="chPw()">Change</button></div>`;
}
function chPw(){
 if($('op').value!=me.p)return msg('Old password is incorrect');
 if($('np').value.length<6)return msg('New password must have at least 6 characters');
 if($('np').value!=$('cp').value)return msg('Passwords do not match');
 me.p=$('np').value;msg('Password changed');render();
}

/* ---------- passenger: search, seats, passengers, payment ---------- */
function cities(){return [...new Set(D.buses.flatMap(b=>[b.from,b.to]))]}
function searchV(){
 if(cmBus)return cmV();
 if(flow)return flowV();
 const c=cities().map(x=>`<option>${x}</option>`).join('');
 let h=`<div class="card"><h2>Search Buses</h2><div class="row">
 <label>From<select id="sf"><option value="">Any</option>${c}</select></label><label>To<select id="st"><option
    value="">Any</option>${c}</select></label>
 <label>Date<input type="date" id="sd" min="${today()}" value="${q.d||add(today(),3)}"></label>
 <label>Time<select id="sp"><option value="">Any
    time</option><option>Morning</option><option>Afternoon</option><option>Evening</option><option>Night</option></select></labe
    l>
 <button class="b" onclick="srch()">Search</button></div></div>`;
 if(q.go){
  const l=D.buses.filter(b=>(!q.f||b.from==q.f)&&(!q.t||b.to==q.t)&&(!q.p||period(b.time)==q.p));
  h+=`<div class="card tw"><h2>Available Buses on
    ${q.d}</h2><table><tr><th>Bus</th><th>Type</th><th>Route</th><th>Time</th><th>Fare</th><th>Seats left</th><th></th></tr>`+
  (l.map(b=>{const left=b.seats-taken(b.id,q.d).length;return `<tr><td>${esc(b.name)}<div
    class="mut">${esc(b.desc)}</div></td><td>${esc(b.cat)}</td><td>${b.from} → ${b.to}</td><td>${b.time}</td><td>Rs.
    ${b.fare}</td>
  <td class="${left?'ok':'bad'}">${left}</td><td><button class="b s" ${left?'':'disabled'} onclick="pick(${b.id})">Select
    Seats</button> <button class="b a s" onclick="cmBus=${b.id};render()">Comments</button></td></tr>`}).join('')||'<tr><td
    colspan="7" class="mut">No buses found.</td></tr>')+'</table></div>';
 }
 return h;
}
function srch(){
 const d=$('sd').value;if(!d||d<today())return msg('Select a valid travel date');
 q={go:1,f:$('sf').value,t:$('st').value,p:$('sp').value,d};
 if(q.f&&q.f==q.t)return msg('From and To cannot be the same');
 render();$('sf').value=q.f;$('st').value=q.t;$('sp').value=q.p;
}
function pick(id){flow={b:id,d:q.d,sel:[],stage:'seats'};render()}
function flowV(){
 const b=bus(flow.b),t=taken(flow.b,flow.d);
 const head=`<div class="card"><h2>${esc(b.name)} (${esc(b.cat)})</h2><div>${b.from} → ${b.to} | ${flow.d} | ${b.time} | Rs.
    ${b.fare} per seat</div></div>`;
 if(flow.stage=='seats'){
  let s='';for(let i=1;i<=b.seats;i++){
   s+=`<button class="seat ${t.includes(i)?'x':flow.sel.includes(i)?'on':''}" ${t.includes(i)?'disabled':''}
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    onclick="tog(${i})">${i}</button>`;
  }
  return head+`<div class="card"><h2>Select Seats (max 6)</h2><div class="seats">${s}</div>
  <div class="mut">Yellow = selected, struck out = booked.</div><p>Selected: <b>${flow.sel.join(', ')||'none'}</b> | Total:
    <b>Rs. ${flow.sel.length*b.fare}</b></p>
  <button class="b" onclick="toPax()">Continue</button> <button class="b a" onclick="go('search')">Back</button></div>`;
 }
 if(flow.stage=='pax')return head+`<div class="card"><h2>Passenger Details</h2>`+flow.sel.map((s,i)=>`<div class="row"><b>Seat
    ${s}</b>
  <label>Name<input id="pn${i}" value="${esc((flow.pax[i]||{}).n||'')}"></label><label>Age<input id="pa${i}" type="number"
    min="1" max="110" style="width:80px" value="${(flow.pax[i]||{}).a||''}"></label></div>`).join('')+
  `<button class="b" onclick="toPay()">Proceed to Payment</button> <button class="b a"
    onclick="flow.stage='seats';render()">Back</button></div>`;
 return head+`<div class="card login" style="max-width:420px"><h2>Secure Payment</h2><p>Amount payable: <b>Rs.
    ${flow.sel.length*b.fare}</b></p>
  <input id="cn" placeholder="Card number (16 digits)" maxlength="19"><input id="ch" placeholder="Name on card"><div
    class="row"><input id="cx" placeholder="MM/YY" style="width:46%"><input id="cv" type="password" placeholder="CVV"
    maxlength="3" style="width:46%"></div>
  <button class="b" onclick="payNow()">Pay Now</button> <button class="b a" onclick="flow.stage='pax';render()">Back</button><p
    class="mut">Simulated payment gateway - no real money is charged.</p></div>`;
}
function tog(i){
 const s=flow.sel,k=s.indexOf(i);
 if(k>=0)s.splice(k,1);else if(s.length>=6)return msg('Maximum 6 seats per booking');else s.push(i);
 flow.sel.sort((a,b)=>a-b);render();
}
function toPax(){if(!flow.sel.length)return msg('Select at least one seat');flow.pax=flow.pax||[];flow.stage='pax';render()}
function toPay(){
 const p=[];
 for(let i=0;i<flow.sel.length;i++){
  const n=$('pn'+i).value.trim(),a=+$('pa'+i).value;
  if(!n||!(a>0))return msg('Enter name and age for every passenger');
  p.push({n,a});
 }
 flow.pax=p;flow.stage='pay';render();
}
function payNow(){
 const cn=$('cn').value.replace(/\s/g,'');
 if(!/^\d{16}$/.test(cn))return msg('Card number must be 16 digits');
 if(!$('ch').value.trim())return msg('Enter the name on the card');
 if(!/^(0[1-9]|1[0-2])\/\d{2}$/.test($('cx').value))return msg('Expiry must be MM/YY');
 if(!/^\d{3}$/.test($('cv').value))return msg('CVV must be 3 digits');
 if(flow.sel.some(s=>taken(flow.b,flow.d).includes(s)))return msg('Sorry, a selected seat was just booked');
 const b=bus(flow.b);
 const bk={id:++nid,u:me.u,bus:flow.b,date:flow.d,seats:flow.sel.slice(),pax:flow.pax,amt:flow.sel.length*b.fare,st:'Booked'};
 D.bk.push(bk);
 msg('Payment successful! Ticket #'+bk.id+' booked. Confirmation email sent to '+me.email+'.');
 go('my');
}
function myV(){
 return `<div class="card tw"><h2>My Booking History</h2><table><tr><th>Ticket</th><th>Bus</th><th>Route</th><th>Date</th><th>Se
    ats</th><th>Passengers</th><th>Amount</th><th>Status</th><th></th></tr>`+
 (D.bk.filter(b=>b.u==me.u).map(b=>`<tr><td>#${b.id}</td><td>${esc(bus(b.bus).name)}</td><td>${bus(b.bus).from} →
    ${bus(b.bus).to}</td><td>${b.date}</td><td>${b.seats.join(', ')}</td>
 <td>${b.pax.map(p=>esc(p.n)+' ('+p.a+')').join(', ')}</td><td>Rs. ${b.amt}</td><td
    class="${b.st=='Booked'?'ok':'bad'}">${b.st}</td>
 <td>${b.st=='Booked'?`<button class="b a s" onclick="cancelT(${b.id})">Cancel
    Ticket</button>`:''}</td></tr>`).join('')||'<tr><td colspan="9" class="mut">No bookings yet.</td></tr>')+'</table></div>';
}
function cancelT(id){
 const b=D.bk.find(x=>x.id==id);
 if(b.date<today())return msg('Past journeys cannot be cancelled');
 if(!confirm('Cancel ticket #'+id+'?'))return;
 b.st='Cancelled';msg('Ticket cancelled. Refund of Rs. '+Math.round(b.amt*0.8)+' (80%) will be credited.');render();
}
function newsV(){return `<div class="card"><h2>News</h2>`+D.news.map(n=>`<p><b>${n.date}</b> -
    ${esc(n.text)}</p>`).join('')+'</div>'}
function cmV(){
 const b=bus(cmBus);
 return `<div class="card"><h2>Comments on ${esc(b.name)}</h2>`+(D.cm.filter(c=>c.bus==cmBus).map(c=>`<p><b>${esc(c.u)}</b>
    <span class="mut">${c.date}</span><br>${esc(c.text)}</p>`).join('')||'<p class="mut">No comments yet.</p>')+
  `<div class="row"><input id="ct" placeholder="Write a comment" style="flex:1"><button class="b"
    onclick="addCm()">Post</button><button class="b a" onclick="cmBus=null;render()">Back</button></div></div>`;
}
function addCm(){
 const t=$('ct').value.trim();if(!t)return msg('Enter a comment');
 D.cm.push({id:++nid,bus:cmBus,u:me.u,text:t,date:today()});render();
}

/* ---------- admin ---------- */
function catsV(){
 return `<div class="card"><h2>Bus Categories</h2><div class="row"><input id="nc" placeholder="New category (e.g.
    Volvo)"><button class="b" onclick="addCat()">Add</button></div>`+
 D.cats.map((c,i)=>`<p>${esc(c)} <button class="b a s" onclick="delCat(${i})">Delete</button></p>`).join('')+'</div>';
}
function addCat(){const c=$('nc').value.trim();if(!c)return msg('Enter category name');if(D.cats.includes(c))return
    msg('Category exists');D.cats.push(c);render()}
function delCat(i){if(D.buses.some(b=>b.cat==D.cats[i]))return msg('Category is in use by a bus');D.cats.splice(i,1);render()}
function busesV(){
 const e=D.buses.find(b=>b.id==editBus)||{name:'',cat:D.cats[0],from:'',to:'',time:'',fare:'',seats:'',desc:'',photo:''};
 return `<div class="card"><h2>${editBus?'Update Bus':'Add Bus'}</h2><div class="row">
 <label>Name<input id="bn" value="${esc(e.name)}"></label><label>Category<select id="bc">${D.cats.map(c=>`<option
    ${c==e.cat?'selected':''}>${esc(c)}</option>`).join('')}</select></label>
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 <label>From<input id="bf" value="${esc(e.from)}"></label><label>To<input id="bt" value="${esc(e.to)}"></label>
 <label>Time<input id="bm" type="time" value="${e.time}"></label><label>Fare (Rs.)<input id="bp" type="number" min="1"
    value="${e.fare}" style="width:90px"></label>
 <label>Seats<input id="bs" type="number" min="1" max="40" value="${e.seats}" style="width:80px"></label>
 <label>Photo URL<input id="bh" value="${esc(e.photo)}"></label><label>Description<input id="bd" value="${esc(e.desc)}"></label>
 <button class="b" onclick="saveBus()">Save</button>${editBus?'<button class="b a"
    onclick="editBus=null;render()">Cancel</button>':''}</div></div>
 <div class="card tw"><h2>Buses</h2><table><tr><th>Photo</th><th>Name</th><th>Category</th><th>Route</th><th>Time</th><th>Fare</
    th><th>Seats</th><th></th></tr>`+
 D.buses.map(b=>`<tr><td>${b.photo?`<img src="${esc(b.photo)}" alt=""
    width="40">`:'■'}</td><td>${esc(b.name)}</td><td>${esc(b.cat)}</td><td>${b.from} → ${b.to}</td><td>${b.time}</td><td>Rs.
    ${b.fare}</td><td>${b.seats}</td>
 <td><button class="b a s" onclick="editBus=${b.id};render()">Edit</button> <button class="b a s"
    onclick="delBus(${b.id})">Delete</button></td></tr>`).join('')+'</table></div>';
}
function saveBus(){
 const o={name:$('bn').value.trim(),cat:$('bc').value,from:$('bf').value.trim(),to:$('bt').value.trim(),time:$('bm').value,fare:
    +$('bp').value,seats:+$('bs').value,desc:$('bd').value.trim(),photo:$('bh').value.trim()};
 if(!o.name||!o.from||!o.to||!o.time||o.fare<1||o.seats<1)return msg('Fill name, route, time, fare and seats');
 if(o.from==o.to)return msg('From and To cannot be the same');
 if(editBus)Object.assign(D.buses.find(b=>b.id==editBus),o);else D.buses.push({id:++nid,...o});
 editBus=null;render();
}
function delBus(id){
 if(D.bk.some(b=>b.bus==id&&b.st=='Booked'&&b.date>=today()))return msg('Bus has upcoming bookings and cannot be deleted');
 D.buses=D.buses.filter(b=>b.id!=id);render();
}
function usersV(){
 return `<div class="card tw"><h2>Registered
    Users</h2><table><tr><th>Username</th><th>Name</th><th>Email</th><th>Phone</th><th></th></tr>`+
 D.users.map((u,i)=>`<tr><td>${esc(u.u)}</td><td>${esc(u.name)}</td><td>${esc(u.email)}</td><td>${esc(u.phone)}</td><td><button
    class="b a s" onclick="D.users.splice(${i},1);render()">Delete</button></td></tr>`).join('')+'</table></div>';
}
function anewsV(){
 return `<div class="card"><h2>Manage News</h2><div class="row"><input id="nt" placeholder="News text" style="flex:1"><button
    class="b" onclick="addNews()">Publish</button></div>`+
 D.news.map((n,i)=>`<p><b>${n.date}</b> - ${esc(n.text)} <button class="b a s"
    onclick="D.news.splice(${i},1);render()">Delete</button></p>`).join('')+'</div>';
}
function addNews(){const t=$('nt').value.trim();if(!t)return msg('Enter news
    text');D.news.unshift({id:++nid,text:t,date:today()});render()}
function acmV(){
 return `<div class="card tw"><h2>User
    Comments</h2><table><tr><th>Bus</th><th>User</th><th>Date</th><th>Comment</th><th></th></tr>`+
 (D.cm.map((c,i)=>`<tr><td>${esc(bus(c.bus).name)}</td><td>${esc(c.u)}</td><td>${c.date}</td><td style="white-
    space:normal">${esc(c.text)}</td><td><button class="b a s"
    onclick="D.cm.splice(${i},1);render()">Delete</button></td></tr>`).join('')||'<tr><td colspan="5" class="mut">No
    comments.</td></tr>')+'</table></div>';
}
function repV(){
 const ok=D.bk.filter(b=>b.st=='Booked'),rev=ok.reduce((s,b)=>s+b.amt,0);
 return `<div class="card"><h2>Report</h2><p>Buses: <b>${D.buses.length}</b> | Users: <b>${D.users.length}</b> | Confirmed
    bookings: <b>${ok.length}</b> | Cancelled: <b>${D.bk.length-ok.length}</b> | Revenue: <b>Rs. ${rev}</b></p></div>
 <div class="card tw"><h2>Bookings per Bus</h2><table><tr><th>Bus</th><th>Route</th><th>Bookings</th><th>Seats
    sold</th><th>Revenue</th></tr>`+
 D.buses.map(b=>{const k=ok.filter(x=>x.bus==b.id);return `<tr><td>${esc(b.name)}</td><td>${b.from} →
    ${b.to}</td><td>${k.length}</td><td>${k.reduce((s,x)=>s+x.seats.length,0)}</td><td>Rs.
    ${k.reduce((s,x)=>s+x.amt,0)}</td></tr>`}).join('')+
 `</table><p><button class="b a" onclick="window.print()">Print Report</button></p></div>`;
}
render();
</script>
</body>
</html>
```

Output

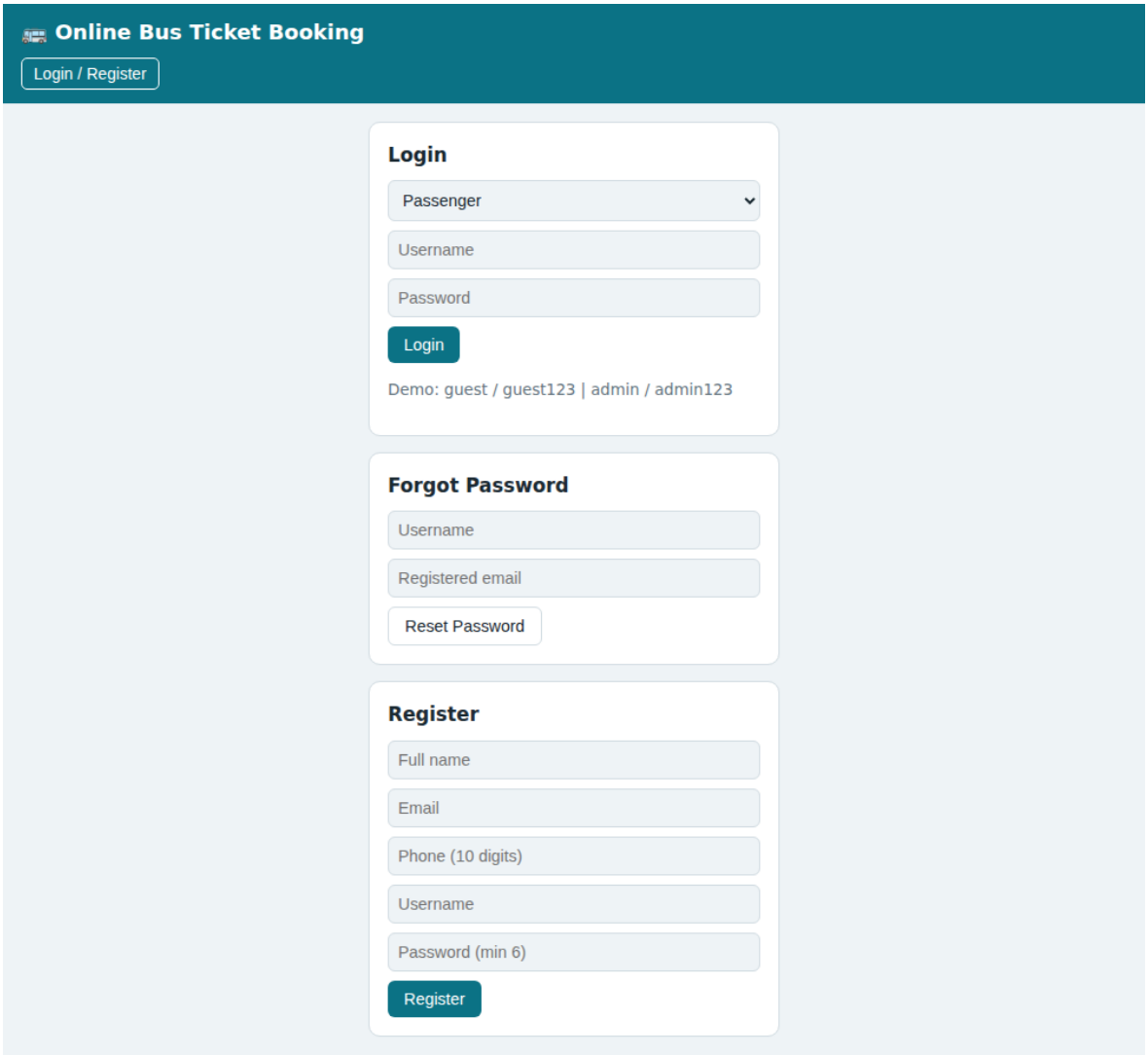


Fig 1: Login, forgot password and registration page

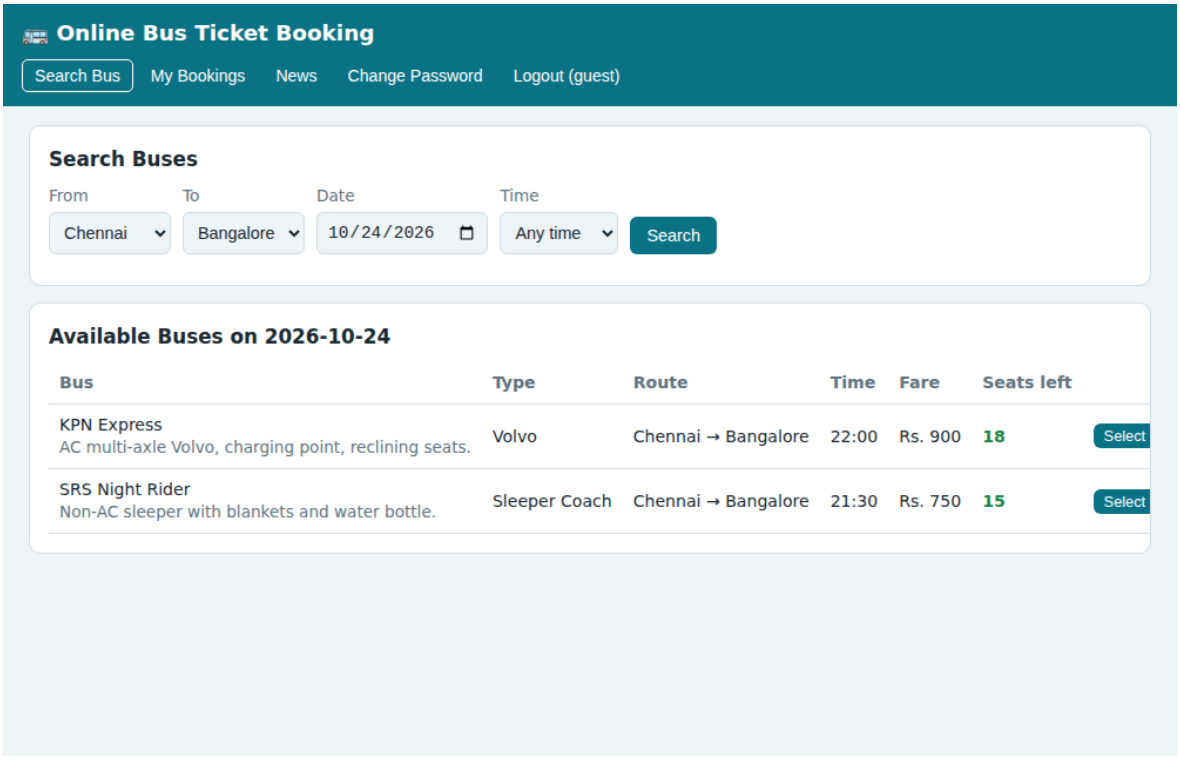


Fig 2: Passenger - Search buses (Chennai to Bangalore)

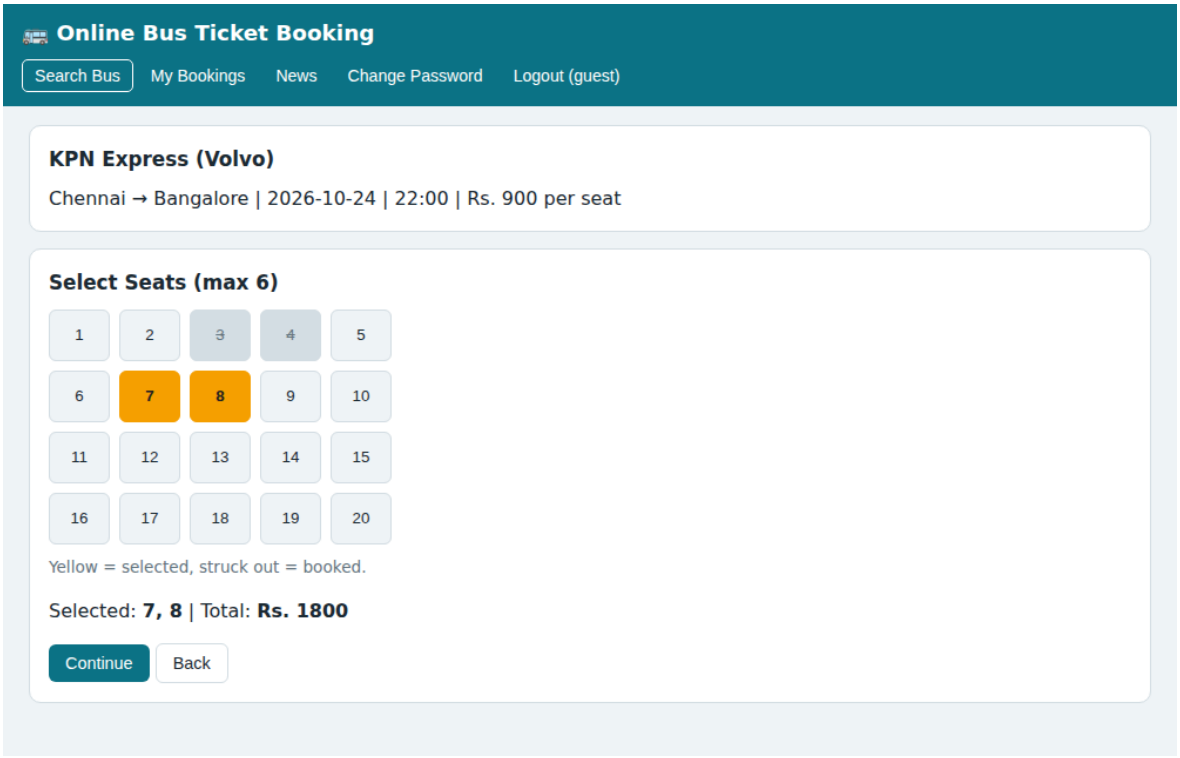


Fig 3: Passenger - Seat selection (booked seats 3 and 4 disabled)

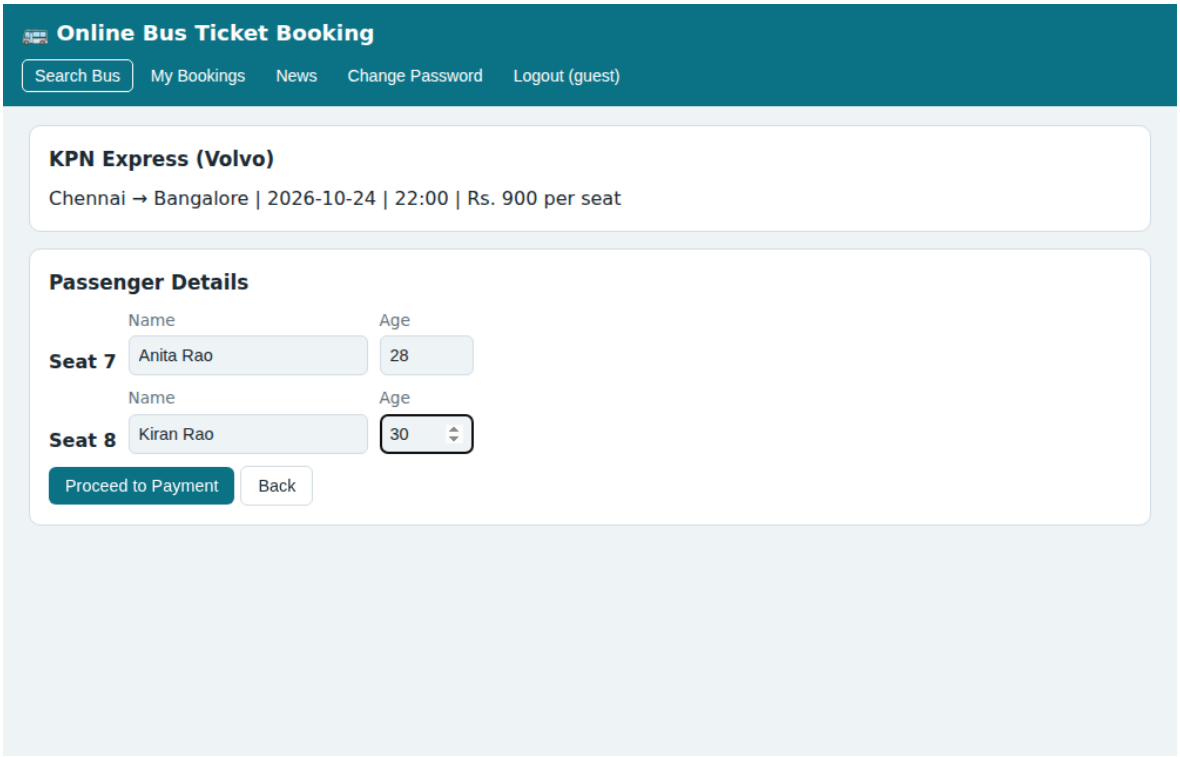


Fig 4: Passenger - Passenger details

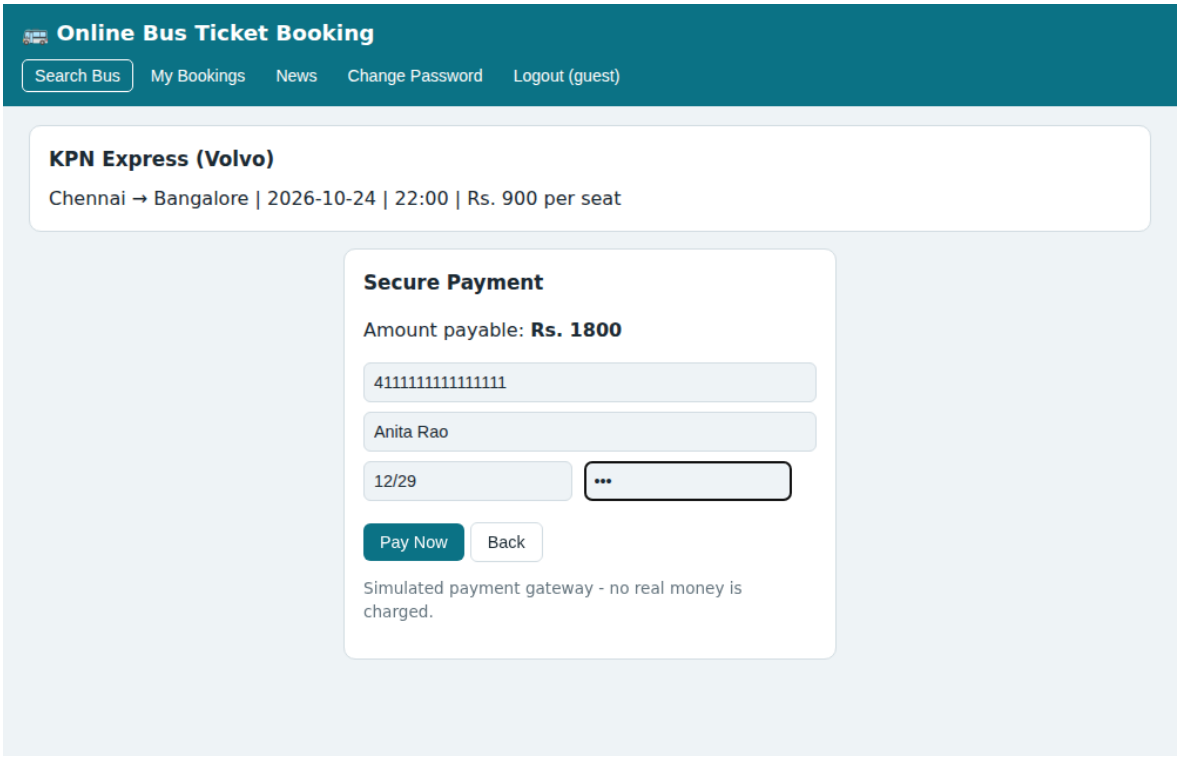


Fig 5: Passenger - Payment gateway

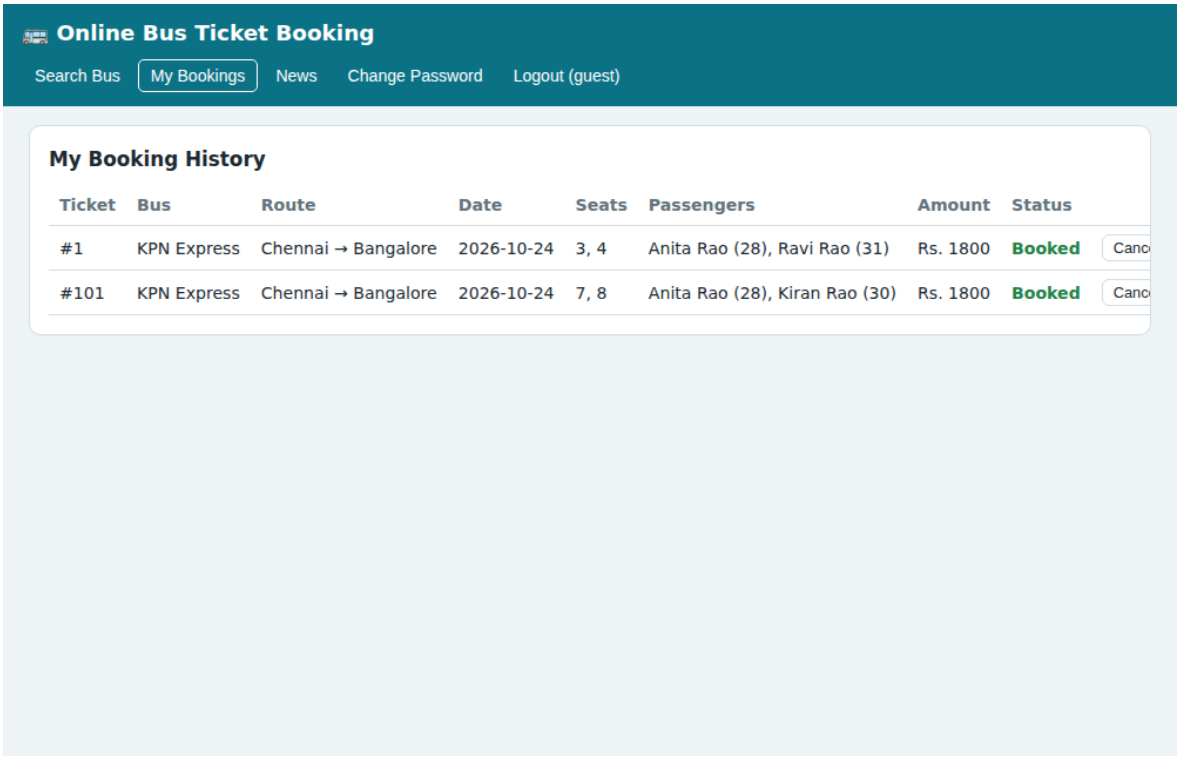


Fig 6: Passenger - Booking history with the new ticket

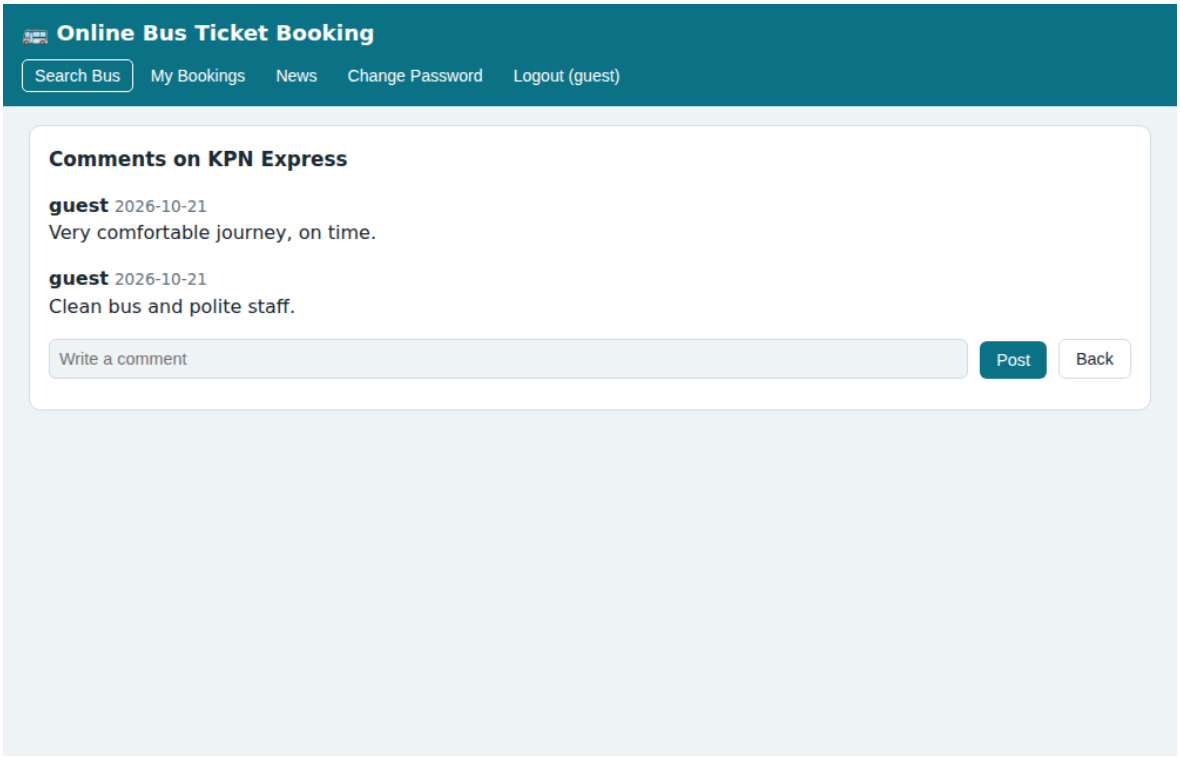


Fig 7: Passenger - Comments on a bus

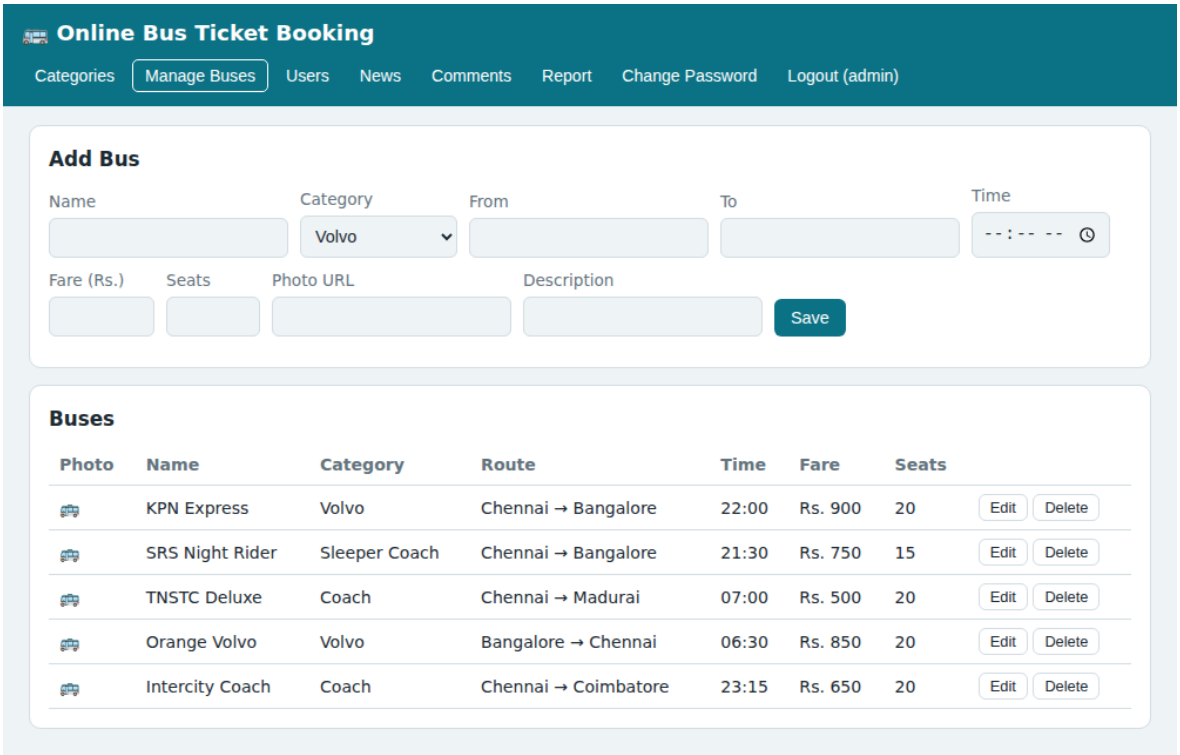


Fig 8: Admin - Manage buses (new bus added)

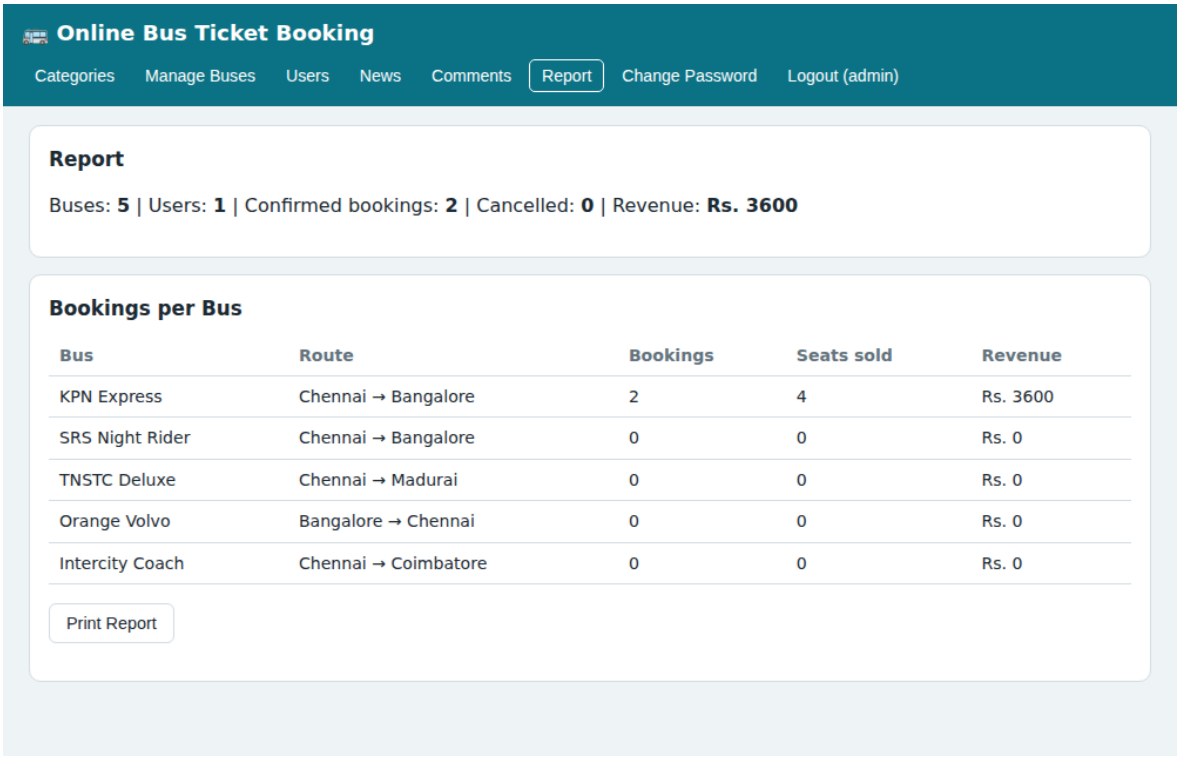


Fig 9: Admin - Report

Result

The online bus ticket booking website was developed using HTML, CSS and JavaScript. Passenger operations (registration, login, password change and reset, search, seat selection, payment, history, cancellation, comments) and administrator operations (categories, bus management, users, news, comments, reports) were executed successfully.